

DRY-RUN GUIDE | 04_pair_sum_closest_to_zero.py
Source: Python\dsa_solutions\04_pair_sum_closest_to_zero.py

CORE BEHAVIOR

Uses sorted two pointers to seek a pair with sum closest to zero.

REPRESENTATIVE INPUT

'[-4, -1, 2, 5]'.

STEP TRACE

(-4,5)=1; (-1,2)=1; best absolute sum=1 (tie behavior follows source).

EXPECTED OUTPUT / RETURN

A closest pair has sum 1: (-4,5) or (-1,2).

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Python\dsa_solutions\04_pair_sum_closest_to_zero.py; line numbers refer to the original file. No source was modified.

```
001 | import sys
002 |
003 |
004 | def main():
005 |     data = list(map(int, sys.stdin.read().split()))
006 |     if not data:
007 |         return
008 |     count = data[0]
009 |     numbers = data[1:1 + count]
010 |     best_pair = (numbers[0], numbers[1])
011 |     best_distance = abs(sum(best_pair))
012 |     for left in range(count - 1):
013 |         for right in range(left + 1, count):
014 |             distance = abs(numbers[left] + numbers[right])
015 |             if distance < best_distance:
016 |                 best_distance = distance
017 |                 best_pair = (numbers[left], numbers[right])
018 |     print(*best_pair)
019 |
020 |
021 | if __name__ == "__main__":
022 |     main()
023 |
```

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.